

Mazen Mohamed Abdelsalam

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Profile

Final-year AI student at Cairo University focused on LLM applications and RAG. Built a LangChain and Groq (Llama-3.3) pytest generator and completed NVIDIA DLI training in generative AI. 2000+ algorithmic problems solved, with top regional ICPC rankings. Seeking an AI engineering internship.

Education

Bachelor of Science in Artificial Intelligence

Sep 2023 – Expected Jun 2027

Faculty of Computers and Artificial Intelligence, Cairo University

- **Department:** Artificial Intelligence
- **GPA: 3.42/4.0** Ranked **20th** out of **170+** students in AI Department
- **Key Courses:** Machine Learning, Data Structures, Database Systems, Software Engineering, Computer Networks, Operating Systems, Computer Architecture, OOP, Web Technologies, Computational Cognitive Science

Obour STEM School

Sep 2020 – Jul 2023

High School Graduate

Projects

Automated Unit Testing Generator

- Built a Python CLI application using LangChain and the Groq API (Llama-3.3) to automatically generate reliable pytest scripts for any Python function.
- Used Python's **ast** library to parse source files locally, ensuring the LLM only processes valid, single-function code blocks to prevent errors and reduce API calls.
- Designed strict system prompts enforcing the Arrange, Act, Assert (AAA) testing standard and correct mocking of external dependencies.
- Created an interactive terminal menu using **pathlib** to automatically save generated test suites in the correct local directory.
- **Tools:** Python, LangChain, Groq API (Llama-3.3), Pytest, Prompt Engineering

HERE Cairo Geospatial Student Hackathon 2026 – Spatial Road Pattern Detection

- Designed and developed a solution to detect spatial patterns in large-scale geospatial datasets under hackathon time constraints.
- Applied feature extraction techniques to identify road networks using probe and Vehicle Probe Data (VPD).
- Processed and cleaned spatial data using GeoPandas and QGIS to enhance dataset quality and usability for downstream analysis.
- **Tools:** Python, Pandas, NumPy, GeoPandas, QGIS

Sorting System – Algorithm Benchmarking Tool

- Implemented **9** sorting algorithms with detailed iteration output and a template-based system supporting **4** data types (int, float, string, char).
- Integrated microsecond-precision runtime benchmarking to compare algorithm performance across varying dataset sizes.
- **Tools:** C++

Achievements

Competitive Programming – ICPC ID

- Solved **2000+** problems in Algorithms and Data Structures; participated in **110+** online contests
- Ranked **814th** / **13,779** in Meta Hacker Cup 2025 – Qualified to Round 2, earned Meta T-Shirt Nov 2025
- Ranked **40th** / **430+** teams in 2025 ICPC Egyptian Collegiate Programming Contest Aug 2025
- Ranked **18th** / **110+** teams in 2024 ICPC Africa and Arab Collegiate Programming Championship Dec 2024
- Ranked **3rd** / **300+** teams in 2024 ECPC Qualifications – Day 5 Jul 2024
- **Silver Honour** – IYMC International Youth Math Challenge Dec 2022
- **Gold Award** – Design Thinking With Robotics and Computational Thinking International Competition (Grade 11/12) Jun 2022

Skills

Programming Languages: Python, C++, SQL, Java, JavaScript, HTML, CSS

Frameworks & Tools: LangChain, Pandas, NumPy, GeoPandas, Matplotlib, Scikit-learn, Pytest, Groq API, QGIS

Concepts: Machine Learning, Large Language Models (LLMs), Prompt Engineering, Retrieval-Augmented Generation (RAG), Data Analysis, Feature Engineering, Algorithms, Data Structures, OOP, Software Engineering, Database Systems

Leadership & Volunteering

Data Structures and Algorithms Coach & Problem Setter

Sep 2024 – May 2025

Faculty Community, Cairo University

- Mentored **100+ students** in data structures and algorithms, preparing them for ECPC and ACPC competitive programming contests.
- Collaborated with a team of problem-setters to author problems with full solutions and test case suites.

Certifications & Training

NVIDIA DLI: Generative AI

Feb 2026

Information Technology Institute (ITI)

Topics: Generative AI, LLMs, RAG, Prompt Engineering, Applied AI Workflows

Huawei HCIA: Artificial Intelligence

Aug 2024 – Sep 2024

Huawei ICT Academy, Cairo University

Topics: Supervised/Unsupervised Machine Learning, Deep Learning Architectures

Competitive Programming Training – Level 2 & 3

Jan 2022 – Sep 2022

Coach Academy

Topics: Graph Theory, Dynamic Programming, String Hashing, Number Theory, Segment Trees, Sparse Tables